SUBJECT: PREDICTIVE ALGORITHM SELECTION: ANSWERING TIME-HORIZON ATTRITION QUERIES

1. Executive Summary

To deliver true business value from our 32-attribute HR dataset, our machine learning models must evolve past simple "Yes/No" predictions. HR leadership requires actionable intelligence regarding *when* an employee will leave, not just *if*. This report evaluates our algorithm options against two critical business questions—Point-in-Time Risk (Q1) and Time-Horizon Risk (Q2)—and recommends a dual-model strategy that integrates seamlessly with our Databricks Dual-Layer Architecture.

2. Defining the Business Questions

Before selecting the technology, we must clearly define what the business is asking:

Question 1 (Point-in-Time Risk): "Given an employee's 32 parameters today, what is their overall probability of attrition?"
Use Case: A general "Flight Risk Score" used to rank the entire workforce from safest to riskiest.

Question 2 (Time-Horizon Risk): "Given the employee is active today, what is the exact probability they leave within 1, 2, 3, 6, and 12 months?"
Use Case: Proactive retention planning. If an employee has a 5% risk at 1 month but an 80% risk at 6 months, HR knows exactly when to intervene.

Question 3 (The "More" / What-If Scenarios): "If we give this employee a 10% raise today, how does their 6-month risk change?" or "Which of our 5 Logical Themes is driving the most 12-month risk?"

3. Algorithm Evaluation: Contenders vs. Rejects

To answer Q1, Q2, and Q3, we must use **Survival Analysis** algorithms. Standard classification algorithms (like basic Random Forest) can only answer Q1.

Why we reject certain algorithms:

- **Kaplan-Meier / Log-Rank Test:** Rejected. These are visual and statistical tools, not predictive models. They cannot ingest 32 parameters to score an individual.

- **Parametric Models (Weibull, Exponential):** Rejected. These force HR data into a strict, smooth mathematical curve. Employee behavior is messy; attrition often happens in sudden "cliffs" (e.g., right after stock options vest). Parametric models miss these cliffs.
- **DeepSurv (Neural Networks):** Rejected. Overkill for 32 tabular columns. It acts as a "black box," making it impossible to explain to HR why an employee is flagged.

4. The Recommended Strategy: A Dual-Model Approach

To answer all business questions with both high accuracy and business trust, I recommend deploying two specific algorithms from our evaluation in tandem.

A. Primary Predictive Engine: Gradient Boosted Survival Models (XGBoost / LightGBM)

This is the absolute best algorithm for tabular HR data. It builds sequential decision trees that learn complex combinations (e.g., High Overtime + Low Satisfaction + Mid-Level).

- **Answers Q1:** Outputs a cumulative hazard score that converts cleanly to a 0-100% General Flight Risk.
- **Answers Q2:** Uses math to "shift" a baseline survival curve for each employee. We can query this curve at exactly months 1, 2, 3, 6, and 12 to get precise time-horizon probabilities.
- **Answers Q3 (What-Ifs):** Because it maps feature interactions, we can simulate the future. We can change an employee's MonthlyIncome feature in the model and instantly see if their 6-month risk drops below our intervention threshold.
- **Architecture Fit:** Natively optimized in Databricks. Trains in seconds on 50,000+ rows.

B. Secondary Explainability Engine: Cox Proportional Hazards (Cox PH)

While XGBoost is highly accurate, it can be difficult to explain its logic to non-technical HR leaders. Cox PH acts as our "glass box" validator.

- **Answers Q1 & Q2:** Calculates a Risk Ratio multiplied against a baseline curve to provide the same 1-12 month probabilities.
- **Answers Q3 (The "Why"):** Cox PH provides exact mathematical coefficients. It is the only algorithm that can definitively answer: "Of the 5 Themes, the Sentiment Theme accounts for 42% of the calculated risk, while Compensation only accounts for 11%."
- **Strategic Value:** If HR pushes back on an XGBoost prediction, we use Cox PH to prove the prediction is mathematically grounded in business logic.

(Note: For future Phase 3 Roadmap items, we will evaluate Multi-State Models, which can predict complex paths like "Probability of Promotion -> followed by Attrition", but this is not required for initial deployment).

5. Integration with the Dual-Layer Architecture

Integrating these algorithms into our existing Part 2 Databricks architecture requires no structural changes to Layer 1. We only upgrade the ML compute in Layer 2.

The Output Workflow:

- **Layer 2 Translation:** The 32 attributes are mathematically prepared (Encoded/Scaled) as previously defined.
- **XGBoost Survival Execution:** The model reads Layer 2 and outputs a distinct probability for t=1, 2, 3, 6, 12 months.
- **The Write-Back:** Instead of pushing one "Risk Score" back to Layer 1, we push five specific columns:

```
Prob_Leave_1M
Prob_Leave_3M
Prob_Leave_6M
Prob_Leave_12M
General_Risk_Score
```

The Dashboard View:

Databricks SQL Dashboards connect to Layer 1. HR executives no longer see a single, static score. They see a trend line for an individual employee, allowing them to prioritize interventions based on urgency (e.g., filtering for all employees with a Prob_Leave_1M > 50%).

6. Conclusion

By selecting XGBoost Survival Models as our predictive engine and Cox PH as our explainability engine, we successfully answer both point-in-time and time-horizon attrition questions. This dual-model approach ensures we deliver highly accurate, time-bound predictions while maintaining the transparency required for HR to take actionable, trusting steps toward employee retention.